



Project 2 — Expense Tracker

Decode Labs Python Programming Internship

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Role: Python Programming Intern

Overview

This project is developed as part of the **Decode Labs Python Programming Internship**. It is a robust command-line application designed to empower users to track, organize, and analyze their daily financial expenses efficiently.

Objective

To build a functional, real-world expense management tool that demonstrates mastery over core Python concepts, including dynamic accumulators, structured data handling with JSON, categorical breakdowns, and comprehensive user input validation.

Key Features

+ Add Expenses

Log individual expenses seamlessly with associated categories and descriptive tags.

⚡ Quick-Add Session

Continuous entry mode for rapidly logging multiple expenses without returning to the menu.



Running Total

Real-time expenditure summary using the dynamic accumulator pattern.



Category Breakdown

Categorized spending visualizer complete with visual progress bars to track financial distribution.



Defensive Coding

Safely intercepts and handles invalid inputs gracefully using robust exception blocks without crashing.



Persistent Data

Automatically saves all financial entries to expenses.json ensuring data persists across program runs.

Technical Concepts Implemented

Accumulator Pattern

While Loop Control & Sentinels

Try/Except Exception Handling

Type Casting & Float Precision

JSON I/O Persistence

Modular Data Structures

How to Run

Follow these quick steps to launch and interact with the Expense Tracker:

1. Ensure **Python 3.x** is installed on your operating system.
2. Open your terminal/command prompt and navigate to the project directory.
3. Execute the script using the terminal command below:

```
$ python expense_tracker.py
```

Follow the interactive menu prompts to record, review, and categorize your expenses.

Future Enhancements

- **Graphical User Interface (GUI):** Implementation of a visual interface using tkinter or PyQt.
- **Budgeting & Alerts:** Inclusion of custom date ranges, monthly spending limits, and automated warning thresholds.
- **Financial Reports:** Export functionality for financial summaries into CSV and formatted PDF reports.
- **Database Integration:** Relational database storage using SQLite for handling high-volume transactions securely.